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Catalytic pyrolysis of polyolefin and multilayer packaging based waste plastics: A pilot scale study

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ABSTRACT

The catalytic pyrolysis of different types of polyolefin and multilayer packaging based plastic wastes in the presence of commercial zeolite catalyst was studied in the batch pilot scale reactor. Different types of multi-layer plastics such as biaxial oriented polypropylene (BOPP), metalized biaxial oriented polypropylene layers (MET/BOPP), poly ethylene terepthalate (PET), metalized polyethylene-terepthalate (MET/PET), PET combined polyethylene (PET/PE) and mixed polyolefin plastic wastes obtained from the municipal corporation were pyrolyzed to determine the oil, gas and char distribution. BOPP based plastic waste exhibited higher oil yield and calorific value (65–70%, 45.14 KJ/g) compared to PET based MLPs (17.8 %, 30 KJ/g) and laminated metalized plastics (13 %, 37 KJ/g). Modifying the feed composition by mixing of polyolefins-based waste plastics with PET based MLPs and BOPP/MET BOPP doubled the liquid yield and notably altered the physicochemical characterization of the resulted pyrolysis oil. Char yield in the polyolefins-based waste is observed to be lesser than the MLP based waste plastics. GC–MS analysis revealed the percentage area of hydrocarbons compounds of the pyrolysis oil obtained from PET based MLP experiments contains high fractions of medium and heavier range hydrocarbons (C11 – C20, C21 – C30). Sulfur content in the oil from different MLPs was measured as below the detection limit. Functional groups of hydrocarbons of oil were analyzed using FT-IR. Solids were characterized for the presence of heavy metals such as Al, Cr, Cu, Co, Pd, and Ni.

1. Introduction

The rapid urbanization and increased population have increased the demand for plastics. In India, about 60 % of the plastics end up as waste after a single use and nearly 43 % of all manufactured plastics used for packaging purposes are mostly of single-use (MNRE, 2014). Flexible plastics are ideal for the packaging purpose due to its lightness and versatility (Horodytska et al., 2018). Polyolefins (PO) contribute to the most significant quantity that includes the low-density polyethylene (LDPE), polypropene (PP) and PET (Horodytska et al., 2018). While polyvinyl chloride (PVC), polystyrene (PS), polyamide (PA), and other special polymers may be used in smaller quantities depending on the need (Andrady and Neal, 2009; Horodytska et al., 2018). Multilayer plastics are extensively preferred in packaging industry because it provides the satisfactory protection and storage of consumer goods while reducing the cost and material requirement compared to the single polymeric plastics (Butler and Morris, 2016). Multi-layered packaging (MLP) often consists of a thin layer of aluminum foil and multiple layers of lamination in paper and plastics which act as barrier to light, moisture, oxygen and carbon dioxide (Jambeck et al., 2015; Morris, 2017; Preston, 1981). MLP plastics also contain various chemical additives such as fillers, plasticizers, flame-retardants, colorants, stabilizers, lubricants, foaming agents, and antistatic agents (Groh et al., 2019). MLP's are also graphics friendly. Owing to the complex composition, MLP waste is technically posing challenges at the large scale treatment processing (Horodytska et al., 2018; Veksha et al., 2020). Thus post-consumer plastics are often end up in the landfills and ocean without any processes (Hopewell et al., 2009).

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Plastic Waste Management Rules of 2016 in India has directed the producers and brand owners to introduce a collect-back system known as the Extended Producers Responsibility (Bhattacharya et al., 2018). Central Pollution Control Board (CPCB) of India estimated the collection efficiency of solid wastes as 80.28 % in 2014, out of which only 28.4 % was treated (Bhattacharya et al., 2018). Most of these plastics are recycled by melting and remolding into lower quality products. However after the successive cycles of melting and remolding, the quality of the polyolefins reduces significantly which require end of life cycle process (Serrano et al., 2012). Incineration is the common alternative process used in many countries to treat the plastic waste and recover energy simultaneous by taking the advantage of the high calorific value segregated plastic wastes (~40 – 44 MJ/kg). However, incineration process possess the high risk of emitting toxic gases such as dioxins and furans (Buekens and Huang, 1998). Furthermore, through gasification process yield synthesis gas, involves high the cost and require large size plants to be of more economical benefits (Vispute et al., 2010).

Pyrolysis is the process of thermal decomposition of plastic polymers into smaller products at elevated temperature in the range of (400–550 °C) in the absence of oxygen to yield liquid oil, gas and char in the presence (catalytic pyrolysis) /absence (thermal pyrolysis) of catalysts (Al-Salem et al., 2017; Sharuddin et al., 2016). Catalysts are widely used to optimize the overall pyrolysis process efficiency by reducing process temperature and time, and improving the quality and quantity of liquid oil through directing specific reactions such as cracking, oligomerization, cyclic, aromatic compounds formation and isomerization reactions (Miandad et al., 2019; Serrano et al., 2012). Thus decreases the overall parasitic energy demand and optimizes the process to attain the economic benefits (Miandad et al., 2016). Variety of catalyst that is extensively used for catalytic pyrolysis are ZSM-5, natural zeolite, Y-zeolite, HZSM-5, FCC, MCM-41, red mud, etc. for the pyrolysis of different plastics such as LDPE, HDPE, PP, PS and waste plastic (Budsaereechai et al., 2019a; Kalargaris et al., 2017a; Kumar et al., 2013; Lee and Shin, 2007; Miandad et al., 2017a, 2017b; Serrano et al., 2012). However, very fewer studies reported the treatment of MLP waste. Few researchers studied the microwave-assisted pyrolysis of MLP plastic waste (Undri et al., 2014; Juliastuti et al., 2017; Lam et al., 2019). Undri et al. (2014) studied the microwave-assisted pyrolysis of multi-layer packaging beverage containing paper, PE, and aluminum. Pyrolysis of paper resulted in the production of bio-oil which containing water, alcohols, aldehydes, acids, and anhydrous sugars and PE resulted in the production of viscous oil containing linear alkanes, alkenes, cyclic, and aromatic hydrocarbons. Juliastuti et al. (2017) used natural zeolite as a catalyst for the microwave pyrolysis of MLP waste-based LDPE and the optimized condition resulted in 28.12 % of liquid oil and 2.88 % of solid. Lam et al. (2019) studied the microwave vacuum pyrolysis for simultaneous conversion of plastic waste and used cooking oil. Most of these studies addressing the MLP plastic processing was carried out through microwave based pyrolysis, which is quite challenging to implement at large scale in the developing world due to the cost associated with the scale up.

Recently Veksha et al. (2020) studied the catalytic pyrolysis of flexible plastic packaging waste and reported the conversion of non-condensable hydrocarbon gases into multi-walled carbon nanotubes. Likewise, catalytic pyrolysis based conversion of PET waste was reported by Du et al. (2016). These studies on the catalytic pyrolysis of MLP waste are done only at the lab scale reactors with the feed capacity of 0.2–100 g. Selpiana et al. (2018) evaluated the thermal pyrolysis of expanded polystyrene and MPL waste mixture in a fixed bed reactor at different feed ratios. Catalytic pyrolysis is significantly important to improve the process efficiency and fine-tune the oil quality. Industrial implementation of the catalytic pyrolysis process is highly influenced when the expensive catalyst are used in huge quantity which may affect the process economy (Ali et al., 2002). Conducting the pyrolysis experiments with the real/complex plastic wastes using a low cost, easily and commercially available zeolite catalyst at the pilot scale reactor in a relatively large feed quantity could provide the useful insights to pyrolysis plant industrial and governmental personals for handling the waste at large scale level. Nevertheless, the catalytic pyrolysis of different MLPs such as biaxial oriented polypropylene (BOPP), metalized biaxial oriented polypropylene layers (MET/BOPP), polyethylene terepthalate (PET), metalized poly ethylene terepthalate (MET/PET), polyethylene terephthalate combined polyethylene (PET/PE) along with mixed plastic wastes at the pilot scale level is not reported yet.

Table 1
The components and their features of pyrolysis reactor.

Reactor components	Features
Dimensions of pyrolysis tank	OD – 25 cm, ID – 22 cm, H – 45 cm
Reactor capacity for feedstock (L)	17.1
Reactor capacity for feedstock (Kg)	2
Dimensions of condenser 1	L – 40 cm, B – 20 cm, H - 32 cm
Dimensions of condenser II	D – 15 cm, H – 30 cm
Maximum temperature	530 °C

In this study, the catalytic pyrolysis of different types of polyolefin and multilayer packaging based plastic wastes in the presence of commercial zeolite catalyst was studied in the batch pilot scale reactor with the feed quantity of 1 kg/batch. The yields of oil, gas, and char were estimated at different feed ratios. The physicochemical and thermal characteristics of pyrolysis oil were determined. Hydrocarbon fractions of liquid oil were analyzed using GC–MS analysis. Char was characterized including for the presence of heavy metals. The composition of gases obtained was also analyzed.

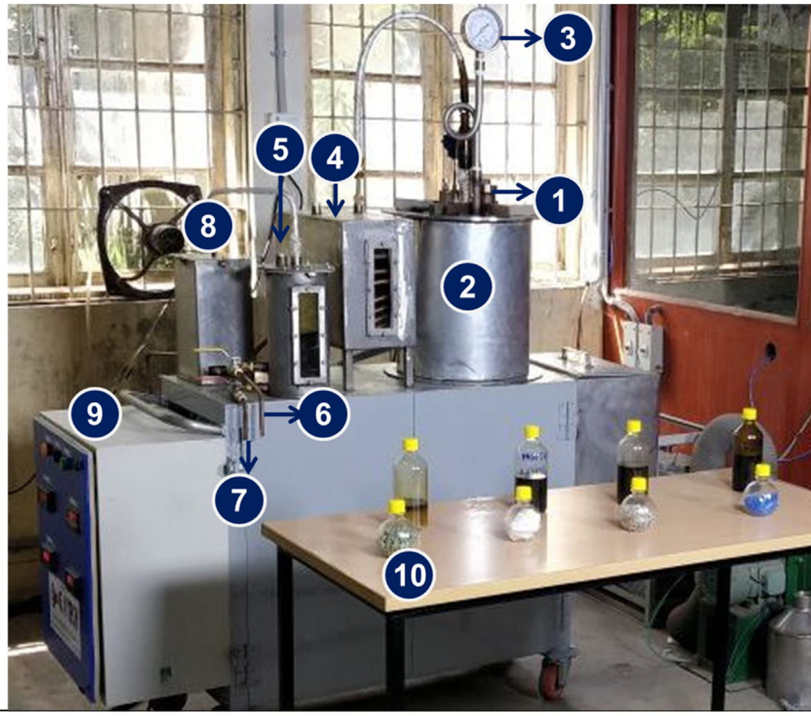
2. Materials and methods

2.1. Materials

Plastic waste samples were collected from different sources. Poly olefin wastes like polyethylene, polypropylene, shredded mixed plastic waste were collected from plastic recyclers and corporation material recovery facility in Chennai. MLP wastes were collected from flexible packaging industries in Chennai. Commercial Ammonium based Zeolite catalyst (SiO_2: Al_2O_3 = 23:1) was used for the batch pyrolysis trials (D.K. enterprises, Chennai).

2.2. Pyrolysis reactor

The photographic image of the pyrolysis reactor used for the experiments are given in Fig. 1 and the details components and their corresponding features are given in the Table 1. A cylindrical stainless steel reactor (diameter – 22 cm, height – 45 cm) was wounded with two electrical heaters in the top and bottom of the inner cylinder to heat the plastics. The inner cylinder was surrounded by an annular cylindrical stainless steel jacket (diameter – 25 cm), which was insulated with glass wool to avoid the heat losses from the pyrolyzer to the surroundings. The temperature measurement was measured by a thermocouple in the top and bottom of the inner cylinder of, and the accuracy of measurement was ±3 °C. The mixer blades were provided for the mixing of plastic and zeolites. The pressure gauge was provided in the pyrolyzer lid to monitor the pressure during the course of the reaction. The heaters, thermocouples were connected with a control panel which has an auto cut-off option at a maximum temperature set point at 500 °C. The cracked hydrocarbon vapor from the waste plastics were passed through the tube side of the condenser, and the chilled water was circulated



Components: 1. Feed Inlet; 2. Pyrolysis chamber; 3. Pressure gauge; 4. Condenser I; 5. Condenser II and oil collection system; 6.Pyrolysis oil collection valve; 7. Water disposal valve; 8. Gas scrubber unit; 9. Controller unit; 10. Pyrolysis oil collected for different types of plastics

Fig. 1. Bench scale plastic pyrolysis plant.

in the shell side of the primary condenser. The water was cooled and circulated through the chiller and water pump arrangement. A cylindrical column of size (diameter – 15 cm, height – 30 cm) filled half with water and the noncondensable gases from the primary condenser is bubbled through it, and pyrolysis oil is collected in the same. Noncondensable gases from the secondary condenser are passed through an activated carbon pack and thermal oxidizer.

The exhaust gases from the thermal oxidizer were connected to a fume hood and chimney arrangement. Shredded plastics and catalysts were homogeneously mixed to effectively crack the long-chain polymer hydrocarbons in to condensable medium-range pyrolysis oil. The resultant product yield (oil, char, and gas) in wt (%) was quantified gravimetrically at the end of each experiment. The experiments were repeated for optimum conditions to maximize the oil yield.

2.3. Pyrolysis experiments

Batch pyrolysis experiments were carried out with 1 Kg of poly olefinic, MLP based plastic waste, and fed into the inner chamber of pyrolyzer, which was surrounded by a stainless steel outer chamber. The temperature range was maintained around 450–500 °C for conducting thermocatalytic depolymerization of different types of plastics. Zeolite beads were used as catalyst material. Zeolite was added into the reactor vessel along with the plastics for insitu cracking process, and the ratio of plastic and zeolite was fixed as 20:1, which was chosen based on previous studies. Mixed plastic wastes were shredded into 2−5 cm size and catalyst used was commercial rod shaped (3 mm diameter and 8 mm length) Zeolite. Plastics and zeolite were mixed and fed in to the pyrolyser for heating. The high viscous molten polymer was mixed by starting the mixer once the temperature in the reactor reaches above 350 °C. The temperature measurement was measured by a thermocouple, and the accuracy of measurement was ±3 °C. The entire system was purged with nitrogen to maintain the inert conditions. The catalytic pyrolysis experiments were carried out with polyolefin-based waste and MLP based plastic wastes in different ratios (Table 2). The resultant product yield (Oil, char, and gas) in wt% was quantified gravimetrically at the end of each experiment. The chemical characterization of the oil products was analyzed by GC–MS (Gas Chromatography-Mass Spectrometer).

2.4. Characterization of plastic wastes and pyrolysis products

The physicochemical, thermal characteristics of oil and char recovered from the poly olefinic and multi-layer packaging (MLP) based waste was analyzed using ASTM methods. The properties of oil like gross calorific value, total sulfur content, kinematic viscosity, density, cetane index, Cleveland flash point of fuel oil recovered from the plastic waste was characterized using ASTM methods (ASTM D48682017, ASTM D48682017, ASTM D42942016e1, ASTM D4452017, ASTM D40522018, ASTM D9762016, ASTM D922016B). The composition of non-condensable gases (CO, CO_2, H_2, and CH_4) evolved during the plastic pyrolysis was analyzed through online gas analyzer (Bhoomi gas analyzer). USEPA 25 A method has been used for analyzing the VOCs present in the flue gases. The gross calorific value of the pyrolysis char was determined through the bomb calorimeter (ASTM D 5865:13). The concentration of heavy metals and sulfur present in the char were determined using ASTM D 6357:2011, ASTM E775 methods. The percentage of polycyclic aromatic hydrocarbons present in the char was determined through GC–MS. The pyrolysis behavior of polyolefin and multi-layer plastics (MLPs) (> 5 mg) was analyzed through the thermogravimetric analyzer (TGA) with 10 °C /min temperature ramp

Table 2
Experimental scheme of catalytic pyrolysis of plastic waste with varying feed and catalyst input.

S. No	Type of plastic waste	Plastic weight (g)	Feed ratio	Catalyst	Catalyst quantity (g)	Retention time (min)	Temperature (^{0}C)
1	LDPE	1000	100	No Catalyst	50	90	450–500
2	LDPE	1000	100	Zeolite	50	90	450–500
3	HDPE	1000	100	Zeolite	50	90	450–500
4	PP	1000	100	Zeolite	50	90	450–500
5	Polyolefinic mixed plastic waste	1000	100	Zeolite	50	90	450–500
6	Polyolefinic mixed plastic waste	1000	100	Zeolite	50	90	450–500
7	Laminaed metallized recycled plastic	1000	100	Zeolite	50	90	450–500
8	PET/PE + Polyolefinic mixed waste	1000	50/50	Zeolite	50	90	450–500
9	PET/MET/PET	1000	100	Zeolite	50	90	450–500
10	PET/MET/PET + Polyolefinic mixed plastic waste	1000	50/50	Zeolite	50	90	450–500
11	BOPP/MET/BOPP	1000	100	Zeolite	50	90	450–500
12	BOPP/MET/BOPP + Polyolefinic mixed plastic waste	1000	60/40	Zeolite	50	90	450–500

rate in the presence of nitrogen gas (Analytical Instruments, India). The change in weight of the sample with reference to temperature was measured. The IR spectra of waste plastic and pyrolysis oil were analyzed using a FT-IR Spectrophotometer (Perkin Elmer) to identify the functional group over the frequency range 4500–600 cm^{-1}. Chemical fingerprinting of oil and char was done through GCMS (Agilent Technologies, USA). A HP-5 (30-m x 0.25-mm) silica-based cross-linked column was used. The injector and detector temperatures were set at 300 °C. The initial temperature was 50 °C held for 1 min, which then elevated to 100 °C with a ramp rate of 10 °C/min and held for 2 min, next ramp rate was 5 °C/min where the temperature was 250 °C held for 2 min; finally, the temperature was elevated to 300 °C with a ramp rate of 3 °C/min held for 15 min. A sample volume of 1 μl was injected in the splitless mode. The carrier gas (Helium) used at the rate of 1 mL/min. MS was scanned from 35 – 550 amu at 1.562 u/s. MS source and quadrupole temperature were maintained at a temperature of 150 °C and 230 °C. Calibration curves were prepared through multiple dilution 20 element multi-elemental total petroleum hydrocarbons (TPH) standard (Supelco, USA). Agilent Technologies Mass Hunter software was used to determine the amount of TPH present in the fuel oil. NIST mass spectral database was used to compare the mass spectra of the unknown organic compounds in the solvent extract.

3. Results and discussion

3.1. Characterization of plastic waste

The thermal behavior of plastics will be useful to identify the decomposition rate at different temperatures. Thermal gravimetric (TG) studies of different type of plastics have been conducted to optimize the temperature and time required for the thermal degradation of waste plastics (Miandad et al., 2019). It also provides the details on the quantity of volatile hydrocarbons and inert materials (fillers and metals) present in the feed plastics to implicate the amount of solid that could be generated at the end of the pyrolysis process. TG curves of PE, PET/PE, and PET/MET/PET is given in Fig. 2. MLP based PET/PE, and PET/MET/PET plastics were compared to PE, since these plastics are considered as representative to the groups of MLP wastes that are not reported in the previous studies (Budsaereechai et al., 2019b; Chattopadhyay et al., 2008; Miandad et al., 2019). For an increase in temperature from 40 to 200 °C, the rate of thermal decomposition was very negligible, and only the softening of plastics has happened. For an increase in temperature from 200 – 350 °C, the weight loss of 90 % and 15 % were observed for polyolefins and MLP due to volatilization, thermal cracking, and vaporization. Increasing the temperature of plastics till 450 °C resulted in complete degradation of polyolefins (99 %) and partial degradation of PET-based MLPs (50 %). The rate of decomposition of MLPs was observed as 70 % till 550 °C. The thermal degradation curve of PET/PE and PET/MET/PET based MLPs were almost stabilized and formed carbon char in the temperature range from 550 to 700 °C. TGA studies clearly shown the necessity of high-temperature requirement for pyrolyzing the MLPs.

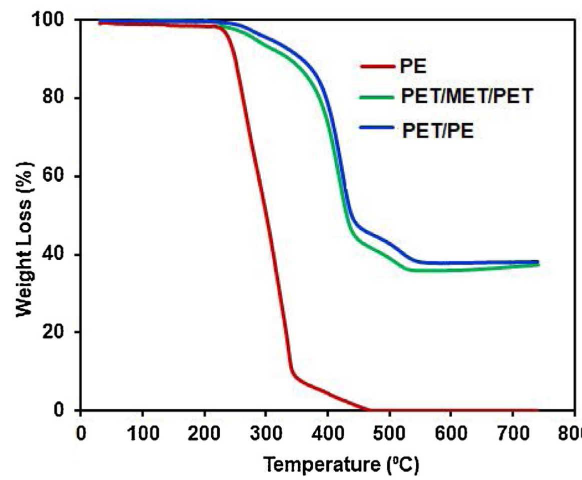


Fig. 2. TGA analysis of Plastic wastes.

3.2. Pyrolysis product yield

Pyrolysis product yield (oil, char, and gas) was estimated based on the weight obtained for the char and oil at the end of the pyrolysis process, and the pyrolysis product yield obtained at different conditions are given in the Fig. 3. The major steps involved in the catalytic reactions during pyrolysis are cracking, oligomerization, cyclic, aromatic compounds formation, and isomerization reactions (Serrano et al., 2012). In the case of thermal pyrolysis (without catalyst), 36.55 wt% and 15 wt% of oil and char yields were obtained, respectively. Pyrolysis of commercial LDPE plastic bags resulted in pyrolysis oil yield of 52 wt%, a gas yield of 38 wt%, and a char yield of 10 wt% for the catalytic pyrolysis process. Pyrolysis of HDPE waste produced 32 wt% of oil, 31 wt% of gas, and 37 wt% of solid. Commercial PP plastic resulted in the production of 53 wt% of oil, 35 wt% of solid, and 12 wt% of gas.

The char yield in polyolefins seems to be high because of the nature of feed stocks used in the trials since they were not virgin. The additives present in the plastics may be formed additional amount of char in the batch pyrolysis trials. The yield values are comparable to the other studies carried out with the waste polyolefinic plastics (Achilias et al., 2007; Miandad et al., 2016; Sriningsih et al., 2014). Pyrolysis of polyolefin-based waste plastics obtained from the Chennai corporation resulted in generating 13 wt% of oil, 17 wt% of gas, and 70 wt% of solid. Solid yield is caused

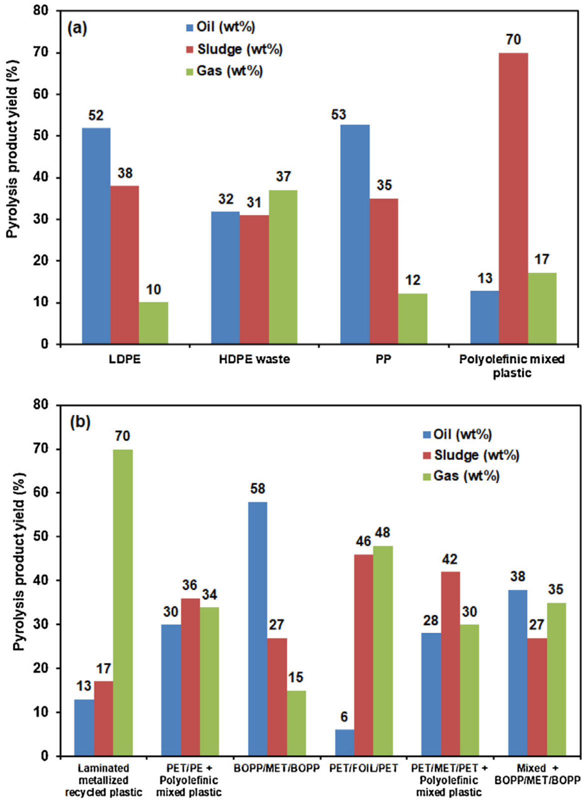


Fig. 3. Pyrolysis products yield (a) Polyolefins waste (b) MLP waste.

due to (i) the presence of high inorganic content in the plastics, (ii) char produced after pyrolysis and (iii) unconverted plastics due to incomplete pyrolysis (Sharma et al., 2014). Apparently, laminated metalized plastic resulted in the highest production of solids of 70 wt% and least oil yield of 13 wt% since most the waste plastics contain a high amount of inorganic contents such as aluminum and zinc metals and filler material.

Different types of multi-layer plastics like biaxial oriented polypropylene (BOPP), metalized biaxial oriented polypropylene layers (MET/BOPP), polyethylene terephthalate (PET), metalized polyethylene terephthalate (MET/PET), polyethylene terephthalate combined polyethylene (PET/PE) and mixed plastic wastes were pyrolyzed. BOPP/MET/BOPP type plastic waste yields 65–70 % oil. BOPP/MET/BOPP (40 %), in combination with mixed plastic waste (60 %) showed around 35–40 % oil yield. Pyrolysis oil yield of PET/MET/PET based MLP is about 17.8 % in the presence of acid catalyst zeolite. Acidic catalyst can promote the cracking and further increase in the cracking promotes the gas formation. Whereas PET based carpet waste in the presence of basic CaO catalysts yielded about ~33 to 37 % (Du et al., 2016). However, undesirable ketones functional groups are also formed due to the basic sites of CaO. Recent study reported by Veksha et al. (2020) on catalytic pyrolysis of PET based flexible MLP using Fe modified ZSM catalyst yielded relatively higher oil yield of 51.2–68.5% shows that modification of catalyst significantly improves the oil yield. However, the cost of catalyst will also increase as per the modification for the large-scale utilization. Further the MLP waste plastics were mixed with other polyolefins-based waste plastic to study the change in pyrolysis products yield. PET/PE and mixed plastic waste yields around 30–40 % oil due to the PE, PP present in mixed plastic waste. PET/MET/PET (50 %) and polyolefinic mixed plastic waste (50 %) yields around 25–30 % oil due to the presence of PE, PP present in the mixed plastic waste. The gas and char composition account for 70–75 % of the PET and METPET based plastic waste. These results represent that the pyrolysis products yield could be notably improved by varying the feed plastic composition.

3.3. Characterization of pyrolysis oil

The physicochemical and thermal characteristics of pyrolysis oil are given in below Table 3. The properties of pyrolysis oil was attained from the MLP wastes, polyolefinic wastes and their combinations in the percentage of 40–60 (that simulates the composition of municipal plastic waste) to understand the resulting pyrolysis oil quality and their corresponding future applications. The density of polyolefin based plastics was almost varied in the 0.78−0.8 g/cc. PET/PE based MLP pyrolysis oil density was higher (0.87 g/cc) when compared to the diesel density of 0.799 g/cc. Sulfur concentration present in the different plastic pyrolysis oil was less than 0.1 %. The viscosity of the pyrolysis oil plays an important role in further applications. The viscosity of the plastic pyrolysis oil obtained from polyolefins and MLPs were varying in the range of (11.7–51.4) Cst and (51.4–275) Cst. The viscosity of diesel should be in the range of (1−4.11) Cst. Flashpoint value of the pyrolysis seem to be lesser than the diesel oil which is comparable to the other studies (Kalargaris et al., 2017b). Other studies in plastic pyrolysis have reported similar results almost 100–200 times higher viscosity values of pyrolysis oil were reported (Mangesh et al., 2020). Based on the results, the pyrolysis oil obtained from the processing of MLPs with mixed plastic waste will be a feasible option to maintain the calorific value of the oil for further applications.

Calorific value is one of the important characteristics based on which the quality of the fuel is evaluated for further applications. Calorific values of pyrolysis oil obtained from various plastics are given in Fig. 4. The calorific values of commercial-grade LDPE were within the range of 40–42 kJ/g. Mixed waste plastics obtained from the municipal corporation was found to be 38–39 kJ/g. The calorific values of HDPE waste plastic and laminated metalized plastic-based pyrolysis oil was observed to be 41 and 39 kJ/g, respectively. The gross calorific value of the pyrolysis oil obtained from different combinations of multi-layer packaging waste varied between 10,789–7156 kcal/kg. BOPP based plastic waste gave higher oil yield and calorific value compared to PET based MLPs. The calorific value of mixed waste plastics obtained from the municipal corporation with MLPs like BOPP and PET was found to be around 45.14 and 30 KJ/g. The calorific values of BOPP and laminated metalized plastic-based pyrolysis oil was observed to be and 30 KJ/g, respectively. These values are found to be similar to those reported for waste plastic pyrolysis oil reported in other studies, which is in the range of 30–47 KJ/g depending on the input plastic quality (Kunwar et al., 2016; Thahir et al., 2019). The lowest calorific value of 30 KJ/g obtained from PET samples can be due to terephthalate formation during the thermal degradation process. The calorific value and density of plastic pyrolysis oil blended with commercial diesel oil has been analyzed. Pyrolysis oil obtained from the mixed plastic waste and MLP (1:1) was blended with the diesel oil in the volumetric ratio of 1:1. The density of the resulting fuel oil was estimated to be 0.78 g/cc. The calorific value of the blended oil was found to be 41.56 MJ/Kg. The resulted calorific value suggests that pyrolysis oil blending with diesel oil could substantially maintain the better calorific value range that can used as fuel oil for boiler heating applications. Kalargaris et al. (2017b) studied the combustion, performance and emission analysis of a DI diesel engine using plastic pyrolysis oil and diesel with different ratios. The lower heating value pyrolysis oil blended with diesel was 38.3 when compared to diesel caloric value of 42.9 MJ/Kg.

The primary reaction pathway of the pyrolysis process involves different steps like initiation, propagation, isomerization, random chain scission and β cleavage, hydrogen transfer, alkylation and aromatization. The strength, density and distribution of acid sites

Table 3
Physico-chemical characteristics of pyrolysis oil obtained from polyolefins and multilayer packaging waste.

S. No	Parameter	Unit	LDPE	BOPP/ MET/BOPP	PET/MET/PET	PET/PE	BOPP/ MET/BOPP + Mixed Plastic	Diesel
1	Carbon	%	100	100	99.4	100	100	100
2	Density	g/cc	0.8	0.78	0.87	0.77	0.79	0.799
3	Flash point	°C	31	31	32	33	32	52–96
4	Gross Calorific Value	KJ/g	42	39	30	40.3	45.14	40–45
5	Sulphur (DL:0.1)	%	BDL	BDL	BDL	BDL	BDL	<0.1–0.6
6	Viscosity	cst	6.8	11.7	51.4	27.5	12.2	1–4.11

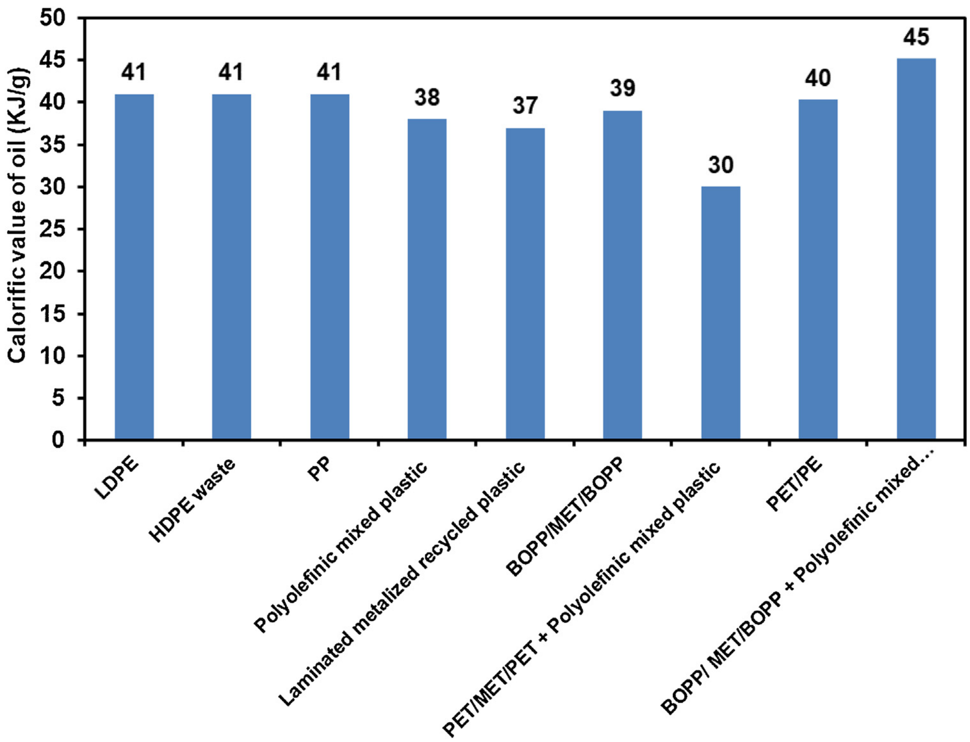


Fig. 4. Calorific values of pyrolysis oil obtained from various plastics.

Table 4
Characterization of char obtained during the pyrolysis of plastic wastes.

Parameter	Unit	LDPE	BOPP/MET/BOPP	PET + PE
Magnesium as Mg	%	17.7	8.7	2.03
Sulphur as SO_3	mg/kg	441	132	34
Aluminum as Al^{+3}	%	1.03	2.54	1.56
Chromium as Cr	mg/kg	37.96	15.38	97.48
Cobalt as Co	mg/kg	4.28	1.0	5.85
Copper as Cu^{+2}	mg/kg	129.09	115.37	213.59
Lead as Pb^{+2}	mg/kg	89.12	217.3	209.91
Nickel as Ni^{+2}	mg/kg	175.41	21.05	164.14
Total Petroleum Hydrocarbon	mg/kg	BLQ(LOQ:0.1)	BLQ(LOQ:0.1)	BLQ(LOQ:0.1)
Poly aromatic Hydrocarbons	mg/kg	BLQ(LOQ:1.0)	BLQ(LOQ:1.0)	BLQ(LOQ:1.0)

present in the catalysts plays a major role in determining the product composition (Panda et al., 2010; Serrano et al., 2005; Aguado et al., 2006; Miskolczi and Bartha, 2008). Secondary reactions are mainly based on intramolecular backbiting with reference to the different position of hydrogen transferred. Mid chain beta scission results in the formation of low molecular weight hydrocarbons. The end chain cyclization of allyl radical results in the formation of cyclic compounds (Zhou et al., 2016). The bond dissociation energy for different C—C bonds present in the polymers were in the order of : Cvinyl–Cvinyl > Cvinyl–Callyl > Calkyl–Calkyl > Calkyl–Callyl > Callyl–Callyl (Vinu and Broadbelt, 2012b). The detailed pyrolysis mechanism for different plastics like polyethylene, polypropylene, poly ethylene terephthalate, polystyrene, etc., have been explained by different research groups (Du et al., 2016; Vinu and Broadbelt, 2012b, Zhoa et al., 2016). The formation mechanism of linear alkenes can be generally attributed to mid-chain b-scission reactions of midchain secondary alkyl radicals. Analysis of pyrolysis oil through GC–MS showed significant production of a large fraction of unsaturated hydrocarbons including linear alkenes (dienes, trienes), and esters. The overlay GC–MS chromatogram of pyrolysis oil obtained from different plastics were shown in Fig. 5 and representative samples were chosen from each category of plastics including polyolefin, MLP and mixed plastics for the GCMS chromatogram. The diesel chromatogram was overlapped

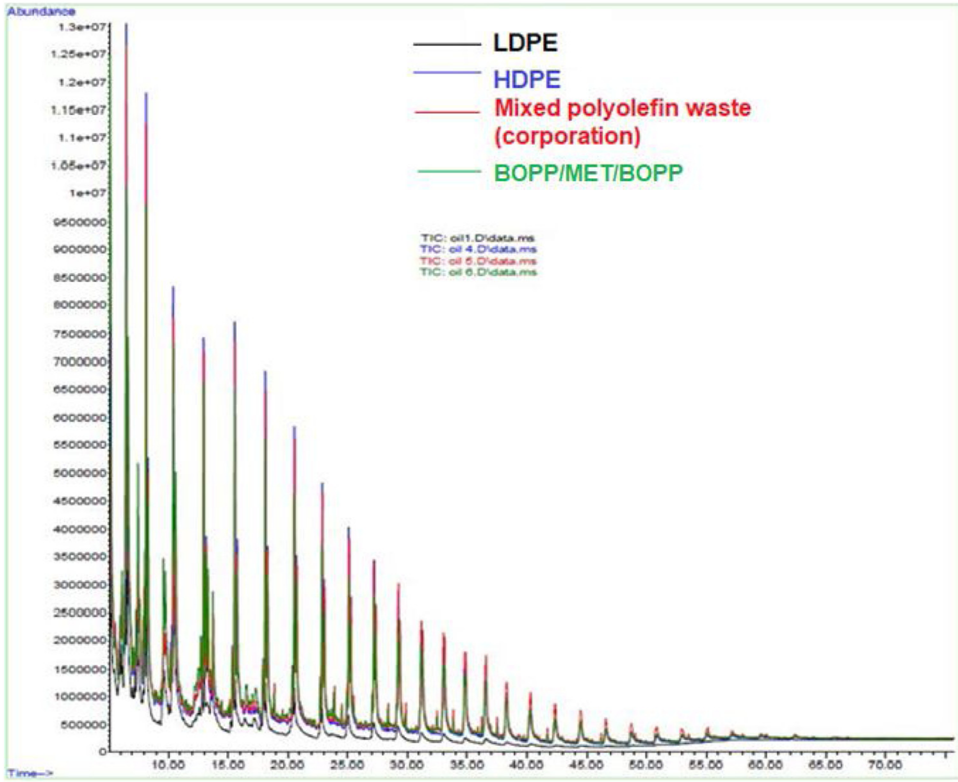


Fig. 5. GC–MS Chromatogram of different pyrolysis oil.

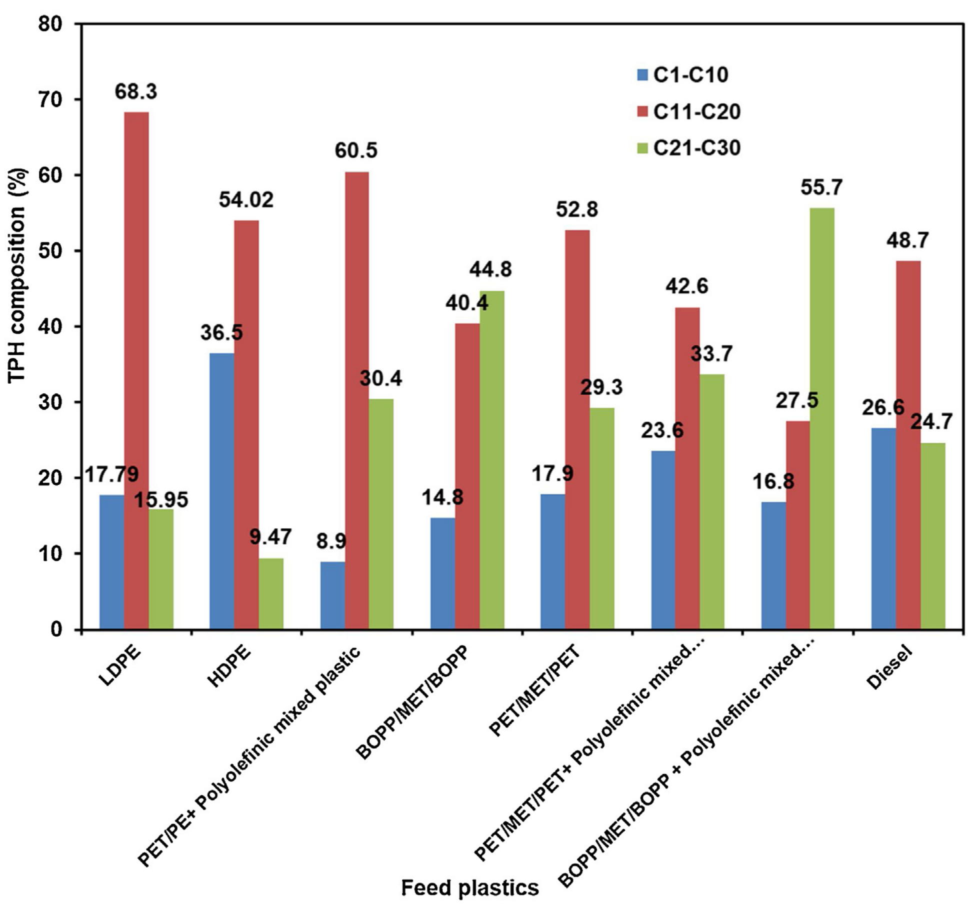


Fig. 6. Variation in TPH distribution for waste plastic with different composition.

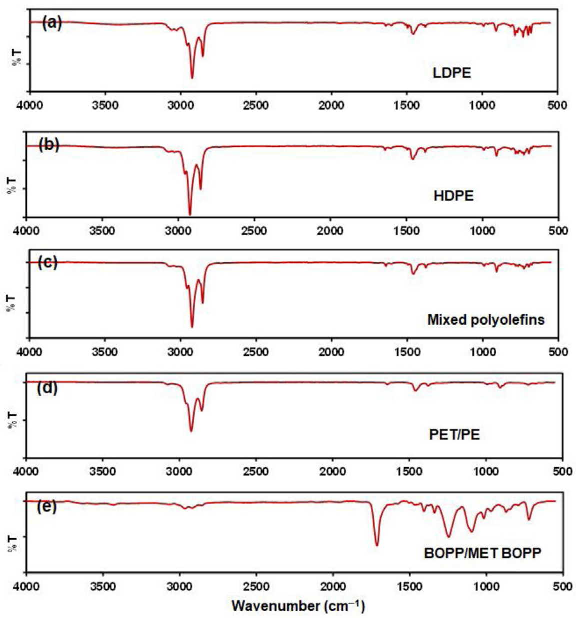


Fig. 7. FTIR analysis of pyrolysis oil.

with pyrolysis oil, and the alkane peaks were almost matched. GC–MS analysis of pyrolysis oil samples with variation in TPH distribution at different feed compositions is shown in Fig. 6. Polycyclic aromatic hydrocarbons like phenol, benzene, naphthalene, and its alkyl derivatives were identified. Above 400–500 °C, there is a sharp increase in aromatic compounds percentage in the oil due to the polymerization of cyclic compounds.

The hydrocarbon range present in the different plastic pyrolysis oils was compared with diesel to evaluate the combustion performance. Due to the presence of paraffinic, olefinic and aromatic HCs in the pyrolysis oil obtained from PE type plastics generally exhibit the high calorific values (Jia et al., 2016; Thahir et al., 2019; Williams and Slaney, 2007). The C5-C10, C11-C20, and C21-C30 carbon numbers present in BOPP/MET/BOPP, PET/MET/PET, PET/MET/PE samples were analyzed from GC–MS chromatogram. The percentage area of (C5-C10) compounds varied in the range of 8.9–23.6 (%) and the medium-range hydrocarbons (C11-C20) varied from 27.5 to 60.5(%). The heavier fractions (C21-C30) varied in the range of 24.7–55.7 (%).

FTIR analysis was carried out to understand the functional groups present in the chemical compounds of pyrolysis oil obtained from different waste plastics (Fig. 7). Chemical bonds of various chemical groups present in the pyrolysis oil produce the infrared absorption spectrum, which assists in understanding the functional groups (Miandad et al., 2017a; Singh et al., 2012).

Representative samples were chosen from each category of plastics including polyolefin, MLP and mixed plastics for the FTIR study. GC–MS analysis result shows that pyrolysis oil obtained from plastics of polyolefins and multi-layer packaging waste contains various hydrocarbons, including aromatics, aliphatics (alkanes and alkenes) along with a minimum of oxygen-containing phenol groups. Most of the peaks obtained for the pyrolysis oil of waste plastic are similar to studies reported by Singh et al. (Singh et al., 2012), Rehan et al. (Rehan et al., 2017), and Miandad et al. (Miandad et al., 2017a). FTIR analysis shows the presence of peaks at 3056, 3075, 3090 $cm^{-1,}$ corresponding to the = C—H stretch of aromatic compounds of hydrocarbons obtained from all of the pyrolysis oil. The presence of sharp peaks observed at 2859, 2928, 2967 cm^{-1} confirms the presence of the C—H stretch of alkanes, which is observed in all of the pyrolysis oils. The presence of peaks observed at 1410, 1470 cm^{-1} belongs to the C-C stretch corresponding to (in-ring) aromatic hydrocarbons. Peaks observed at 916, 1017 cm^{-1} confirms the =C—H bend of alkenes. Peaks corresponding to the aromatic hydrocarbons of C—H were found at 697, 735, 766 cm^{-1} which represents the aromatization capability of the zeolite catalyst (Miandad et al., 2017a).

3.4. Analysis of non-condensable gases from pyrolyzer

The composition of non-condensable gases from the secondary condenser has been collected in tedlar gas sampling bags and analyzed for composition. The composition of gases were measured as CO_2 – 15.8 wt (%), CO – 5014 ppm, benzene – 35.28 ppm, toluene – 8.61 ppm, chlorine – 1.17 ppm and tetrachloroethane – 0.34 ppm. Non condensable gases were adsorbed through carbon adsorption and oxidized in a thermal oxidizer system was provided to burn the off-gases. Thermally oxidized off gases has been analyzed using flue gas analyzer. The composition of exhaust gases in the chimney outlet were measured as CO_2 concentration of 2–5 ppm and CO concentration of 2125–3010 ppm. Bagri and Williams (2002) studied the composition of non-condensable gases generated from different plastics like LDPE, HDPE and PP from catalytic pyrolysis. The exhaust gases were mainly alkanes, with lower concentrations of the alkenes, ethane, propane, and butane. The gas composition in wt (%) were measured as methane – (1.9 %, 1.14 %, 0.93 %), ethane – (2.21 %, 1.67 %, 1.45 %), propane – (1.31, 1.33, 1), propene – (4.56 %, 4%, 3.53 %) and C6 – C12 compounds were in the range of (12–18%, 2–3%, 15–18%). Benzene and toluene concentration measured in the batch pyrolysis trials were similar in the range of reported literature (Bagri and Williams, 2002; Ratnasari et al., 2017). Catalyst plays a major role in increasing the selectivity of product, improving conversion, fuel quality, gas composition and lowering the pyrolysis temperature and residence time. Mesoporous catalysts with high acidity and BET surface area favour higher gases yields, while microporous catalysts with low BET surface area and acidity promote higher liquid oil and char yields (Lopez et al., 2011). Zeaiter (2014) reported 97 % of gases yield from PE catalytic pyrolysis with highly acidic synthetic beta zeolite catalyst. At higher temperatures, intramolecular hydrogen transfer and end chain cyclization of allyl radicals result in the formation of aromatic hydrocarbons (Zhoa et al., 2016). Du et al., 2016 reported that the deoxygenating reactions of synthetic polymers (PET based MLPs) by the catalyst ZSM-5 and CaO. The shape, size and acidic characteristics of ZSM-5 are known to form benzene and other aromatic hydrocarbons in the exhaust gases (Elordi et al., 2009a, 2009b, Dorado et al., 2014; Vasile et al., 2001). Grause et al., 2011 and Kumagai et al., 2011 reported that usage of calcium oxide with the catalyst may be used to improve the pyrolysis oil quality. The presence of VOCs in the non-condensable gases was mainly due to the presence of PET based MLPs present in the feed.

3.5. Characterization of pyrolysis char

The char obtained from plastic pyrolysis using different plastic wastes like LDPE, Biaxial Oriented Poly Propylene (BOPP), and PET/PE based char were characterized (Table 4). The pyrolysis char obtained from different batch experiments varied in the range of 8 – 70 (%) depending upon the type of plastic used. Char yield from polyolefin-based plastics (LDPE, PP, BOPP) varies in the range 8 – 20 %. The additives added in the HDPE based waste increased the char quantity to 30 %. Multi-layer packaging based waste yields 15 – 48 % of char. PET/MET/PET combination and laminated metalized recycled plastic exhibited the maximum amount of char in the range 50 – 70 %. The char obtained from plastic pyrolysis using different plastic wastes like LDPE, Biaxial Oriented Poly Propylene

(BOPP), and PET/PE based char were characterized. The heavy metals concentrations (mg/kg) were analysed and found to be in the range of viz., chromium (15.36 – 97.48), aluminium (1.03 – 2.54), cobalt (1.0 – 5.85), copper (115.37 – 213.59), lead (89.12 – 217.3) and nickel (21.05 – 175.41) respectively. Sulfur as (SO_4^{2-}) concentrations in the char varied from 34 – 441 mg/kg. The input quality of waste plastic determines the quality and quantity of desirable products. The percentage of char and noncondensable gas formation is inversely proportional to the amount of oil produced from plastic waste. The percentage of char formation seems too high for polyolefins and MLPs in the batch system due to heat losses, and it may be considerably reduced in the full-scale plant.

4. Conclusion

An experimental investigation on catalytic pyrolysis of polyolefins and multi-layer plastic (MLP) waste samples were done. The percentage of oil, gas, and char obtained from different plastics has been analyzed. The physicochemical characterization of pyrolysis oil has been done and compared with diesel. The calorific value of oil derived from different combination of polyolefin and multilayer packaging waste varied between 30–47 KJ/g depending on the plastic quality. Pyrolysis oil obtained from PET based MLPs and laminated metalized recycled plastics exhibits lesser yield and low calorific value compared to the polyolefins-based waste plastics (PE, PP and BOPP based MLPs). The physicochemical characteristics such as viscosity and, density of pyrolysis oil obtained from PET based MLPs are 50–100 times higher than the diesel characteristics. Changing the feed composition by mixing of polyolefins-based waste plastics with PET based MLPs and BOPP/MET/BOPP in the range of 40–60 wt% doubled the liquid yield and also notable change in the hydrocarbon fractions was observed based on GC–MS analysis. Sulfur content present in different pyrolysis oil has been determined as 0.1 mg/L, which makes the pyrolysis oil suitable for heating applications. Hydrocarbon fingerprinting of pyrolysis oil and diesel has been matched using GC–MS to understand the feasibility of using it for boiler applications. Heavy metals present in the polyolefinic and MLP pyrolysis char samples have been analyzed for its commercial usage. The catalytic pyrolysis has resulted in volume reduction by 80–90%, along with fuel recovery 50–70%. This study shows that a combination of MLP with polyolefin-based mixed plastic waste will be a feasible solution to obtain a consistent pyrolysis oil quality and quantity. Detailed combustion studies are required for its usage in boiler fuel applications.

Declaration of Competing Interest

The authors report no declarations of interest.

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